

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET
PROVIQEQ

Q.1. Solve the following:

(6x5=30)

- I. Find natural domain of the function, $f(x) = \sqrt{x^2 - 5x + 6}$?
- II. Draw the graph of the equation $y = 3 - (x + 4)^2$?
- III. Find all critical points of $f(x) = x^4(e^x - 3)$?
- IV. Find the two x -intercepts of the function $f(x) = x^2 - 5x + 4$ and confirm that $f'(c) = 0$ at some point c between those intercepts.
- V. Evaluate the integral $\int \frac{\sin(\frac{1}{x}) dx}{3x^2}$?
- VI. Given the function. Compute the limit of the function at $y = -2$?

$$g(y) = \begin{cases} y^2 + 5 & \text{if } y < -2 \\ 1 - 3y & \text{if } y \geq -2 \end{cases}$$

Solve the following.

(3x10=30)

Q.2/a) Find $f'(x)$ using chain rule when $f(x) = [x + \csc(x^3 + 3)]^{-3}$?

$$f(x) = \begin{cases} x^2 + 1 & x \leq 1 \\ 2x & x > 1 \end{cases}$$

- Q.3.a) Use any method to find the relative extrema of the function $f(x) = |3x - x^2|$.
- b) Let L_1 and L_2 be the lines whose parametric equations are $L_1: x = 1 + 2t, y = 2 - t, z = 4 - 2t$ $L_2: x = 9 + t, y = 5 + 3t, z = -4 - t$. Show that L_1 and L_2 intersect at the point $(7, -1, -2)$?
- c) Sketch the region enclosed by the curves $y = 2 + |x - 1|$ and $y = -\frac{1}{5}x + 7$ and find its area using definite integrals.



Solve the following.

Q.x6 Phthd'(x) using chain

B"Detamiae whetherf(x) is continuous and differentiable at $x = 1$

(ii) Given $x^3 + y^3 = 3xy^2$. Find $\frac{dy}{dx}$ by implicit

(iii) Let $y = (\ln x)^{\tan x}$. Find $\frac{dy}{dx}$. ✓

(iv) Evaluate the integral $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$.

(v) Evaluate the integral $\int \sin^4 x \cos^3 x dx$.

(vi) Find an equation of the plane that passes $C(1, 0, -1)$. Pg # 219

0.1. Solve the following questions.
(6/6•30)

(i) the limit $\lim_{x \rightarrow \infty} -z^{-K}$ •
differentiation.

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that passes through the
points $A(-2, 1, 1)$, $B(0, 2, 3)$,

Solve the following questions.

(3x10=30)

Q.1. Int $f(x) = 12x^3$. Find the relative extrema using both first and second derivative

Evaluate the improper integral $\int_{-2}^2 \frac{dx}{x^2}$ Q.2. — if it converges.

Show that the lines $L_1 : x = 2 - t, y = 5 - 2t$ are parallel, and find the distance between them. Q.3. $z = 1 + t$ and $L_2 : x = 1 + 2t, y = 3 - t$

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / First Semester — Spring 2022 Roll No. O.D. 9.d.4 :
Calculus and Analytical Geometry Paper: MS-152 A Time: 3 Hrs. Marks: Subject:

ATTEMPT

Q. 1. f. Solve the following: (6X5=30)

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(i) Evaluate the limit $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 8}$.

(ii) Let $y = x^5 \sec\left(\frac{1}{x}\right)$. Find $\frac{dy}{dx}$.

(iii) that the kymul • $(5 + Az)^n z + 75Az$ is the local approximation of

(iv) the between of $f(x)$ and the a, b (2).
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(v) Evaluate the integral $\int_0^{\sqrt{\pi}} 5x \cos(x^2) dx$.

(vi) Determine whether the points $P_1(6, 9, 7)$, $P_2(9, 2, 0)$, and $P_3(0, -5, -3)$ lie on the same line.

Solve the following.

(3x10=30)

Find: (a) the intervals on which $f(x) = \frac{x}{x^3 + 2}$ is increasing, (b) the intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d) the open intervals on which f is concave down, and (e) the x -coordinates of all inflection points.

Evaluate the integral $\int_0^3 \frac{x^3}{(3 + x^2)^{5/2}} dx$.

Q.2. Q.8.

QA

Find the distance between the skew lines $L_1 : x = 3 - t, y = 4 + 4t, z = 1 + t$ and $L_2 : x = t, y = 3, z = 2t$.

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$$\begin{aligned}
 & \frac{u \cdot 2u + u^2}{u^2 - (u^2 + 2)} \\
 & \frac{2u^2 + u^2}{u^2 - u^2 - 2} \\
 & \frac{3u^2}{-2} \\
 & -\frac{3u^2}{2} \\
 & \frac{1}{2} \int u^3 \cdot \frac{2u}{(3+u^2)^{5/2}} du \\
 & \frac{1}{2} \int u^3 \ln(3+u^2)^{1/2} \\
 & \frac{1}{2} \int u^2 du + \frac{1}{2} \int \ln(3+u^2)^{1/2} \\
 & L. \frac{u^3}{4}
 \end{aligned}$$

